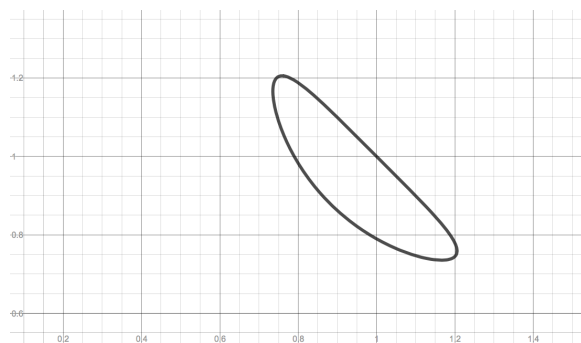


Problem 3

Problem 3

Problem 1 A part of the curve with equation $\cos(\pi xy) + x + y = 1$ is sketched below.



- (a) Use the implicit differentiation to find the derivative dy/dx .
- (b) Consider the point $(1, 1)$. Show (algebraically) that this point lies on the curve.
- (c) Find the equation of the line tangent to the curve at $(1, 1)$. Draw this line in the figure above.